



Convex Optimization

Project 1

spring 1403
Due date: 7th of Ordibehesht



Metric learning. We consider a set of N items, labeled $i = 1, \dots, N$, each one associated with a feature vector $x_i \in \mathbf{R}^n$. We are given data that tells us that some pairs of items are similar, and some pairs are dissimilar. These are given as two sets of pairs of indices, S and D , where $\{i, j\} \in S$ means that items i and j are similar, and $\{i, j\} \in D$ means that items i and j are dissimilar. You can assume that $S \cap D = \emptyset$, i.e., no pairs are both similar and dissimilar, and you can assume that the x_i are distinct. We consider a quadratic metric of the form $d(x, y) = (x - y)^T P (x - y)$, where $P \in \mathbf{S}_+^n$. The matrix P parametrizes the metric. Roughly speaking, we wish to choose P so that $d(x_i, x_j)$ is small when $\{i, j\} \in S$, and large when $\{i, j\} \in D$. We consider one simple formulation of this problem. We create a loss function of the form

$$L(P) = \sum_{\{i,j\} \in S} d(x_i, x_j) + \sum_{\{i,j\} \in D} \frac{1}{d(x_i, x_j)}.$$

(In the righthand sum you can interpret $1/d(x_i, x_j)$ as ∞ if $d(x_i, x_j) = 0$.) We find (or learn) P by minimizing this loss, subject to $P \succeq 0$. The learned metric matrix P can be used to classify items with feature vectors x and y as similar, if $d(x, y) \leq 1$, and dissimilar if $d(x, y) > 1$. The error rate is the fraction of pairs in $S \cup D$ that are misclassified, i.e., $\{i, j\} \in S$ but $d(x_i, x_j) > 1$ or $\{i, j\} \in D$ but $d(x_i, x_j) \leq 1$.

(a) Explain how to find P using convex optimization. Justify any change of variables or other problem transformations.

(b) Carry out the method of part (a) on the data found in `metric_learning_data.py`. Report the optimal value of the objective.

(c) Report the error rate of the learned metric on the training data S and D , and also on a set of test data S^{test} and D^{test} given in the data file.

Good Luck!